Replicating Summary Employment Statistics from Jardim et al. (2022)

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1 The Original Paper

Jardim, Ekaterina, Mark C. Long, Robert Plotnick, Emma van Inwegen, Jacob Vigdor, and Hilary Wething. 2022. "Minimum Wage Increases and Low-Wage Employment: Evidence from Seattle." *American Economic Journal: Economic Policy* 14(2): 263–314.

The paper evaluates the employment and wage effects of Seattle's Minimum Wage Ordinance, which raised the minimum wage from $9.47 to $11 per hour in April 2015 and to $13 per hour in January 2016. The authors use administrative unemployment insurance records from Washington State to directly observe hourly wages for the entire workforce, allowing them to focus precisely on low-wage jobs across all industries. Their empirical strategy combines simple first-difference analysis with synthetic control and interactive fixed effects estimators to isolate the causal effect of the ordinance from Seattle's broader economic boom.

2 The Result of Interest

I replicate the summary employment statistics reported in Table 5 of Jardim et al. (2022), which documents changes in jobs, hours worked, and average wages for Seattle's locatable establishments between 2014 Q2 and 2016 Q2. The paper reports large declines in low-wage jobs and hours worked over this period, even as overall employment expanded dramatically. This result establishes the foundation for the paper's causal synthetic control analysis.

3 Data and Methods

The data come from the publicly available replication package deposited by Jardim et al. (2022) on the ICPSR repository (project 133921). I use the file `region_level_cumulative.dta`, a PUMA-by-quarter aggregated dataset constructed from confidential unemployment insurance records. Each row represents one Public Use Microdata Area in Washington State for one quarter, covering 2014 Q2 through 2016 Q3.

The preprocessing script reads the raw data file, filters to the relevant quarter range, flags Seattle PUMAs using the region indicator variable, and aggregates across PUMAs by summing within each Seattle-by-quarter cell. Average hourly wage rates are derived by dividing total quarterly payroll by total quarterly hours. The cleaned dataset is saved to `temp/clean_data.csv`.

The analysis script reads the cleaned data, extracts the baseline (2014 Q2) and endpoint (2016 Q2) observations for Seattle, and computes the absolute and percentage changes in jobs, hours, and wages for jobs paying under $19 per hour. These values are written directly to a LaTeX table fragment in `output/tables/main_result.tex`, which is included below.

4 Replication Results

Table 1: Replication of Summary Employment Statistics, from Jardim et al. (2022)

	2014 Q2	2016 Q2	Change
Jobs paying < $19/hr	90757	89188	-1.7%
Hours worked < $19/hr	36450720	35466512	-2.7%
Avg. wage < $19/hr	$14.19	$15.00	5.7%

Our replication results are shown in the table above. The direction of all three findings matches the paper exactly — low-wage jobs and hours contracted while wages rose — but the magnitudes are somewhat smaller than those reported in the NBER working paper version (2017). This discrepancy most likely reflects the fact that the publicly available dataset corresponds to the revised 2022 published version, which uses a refined sample compared to the 2017 draft.

5 What I Learned

The most straightforward aspect of this replication was the analysis step itself. Once the data were properly loaded and aggregated, computing percentage changes across two quarters was simple. The harder part was understanding the data structure. The publicly available dataset is already aggregated to the PUMA-by-quarter level, meaning the most computationally intensive work had already been done by the original authors and cannot be fully inspected by an outside replicator. This is an important lesson about replication packages: what is provided is often a processed intermediate file rather than true raw data.

Doing this replication also clarified why the paper's results differ from prior literature. The ability to directly observe hourly wages means the treatment group is precisely defined. A replicator working only with public data like the Quarterly Census of Employment and Wages could not reproduce even this descriptive table. That data limitation is the central methodological contribution of the paper.

References

[1] Jardim, Ekaterina, Mark C. Long, Robert Plotnick, Emma van Inwegen, Jacob Vigdor, and Hilary Wething. 2022. "Minimum Wage Increases and Low-Wage Employment: Evidence from Seattle." *American Economic Journal: Economic Policy* 14(2): 263–314.